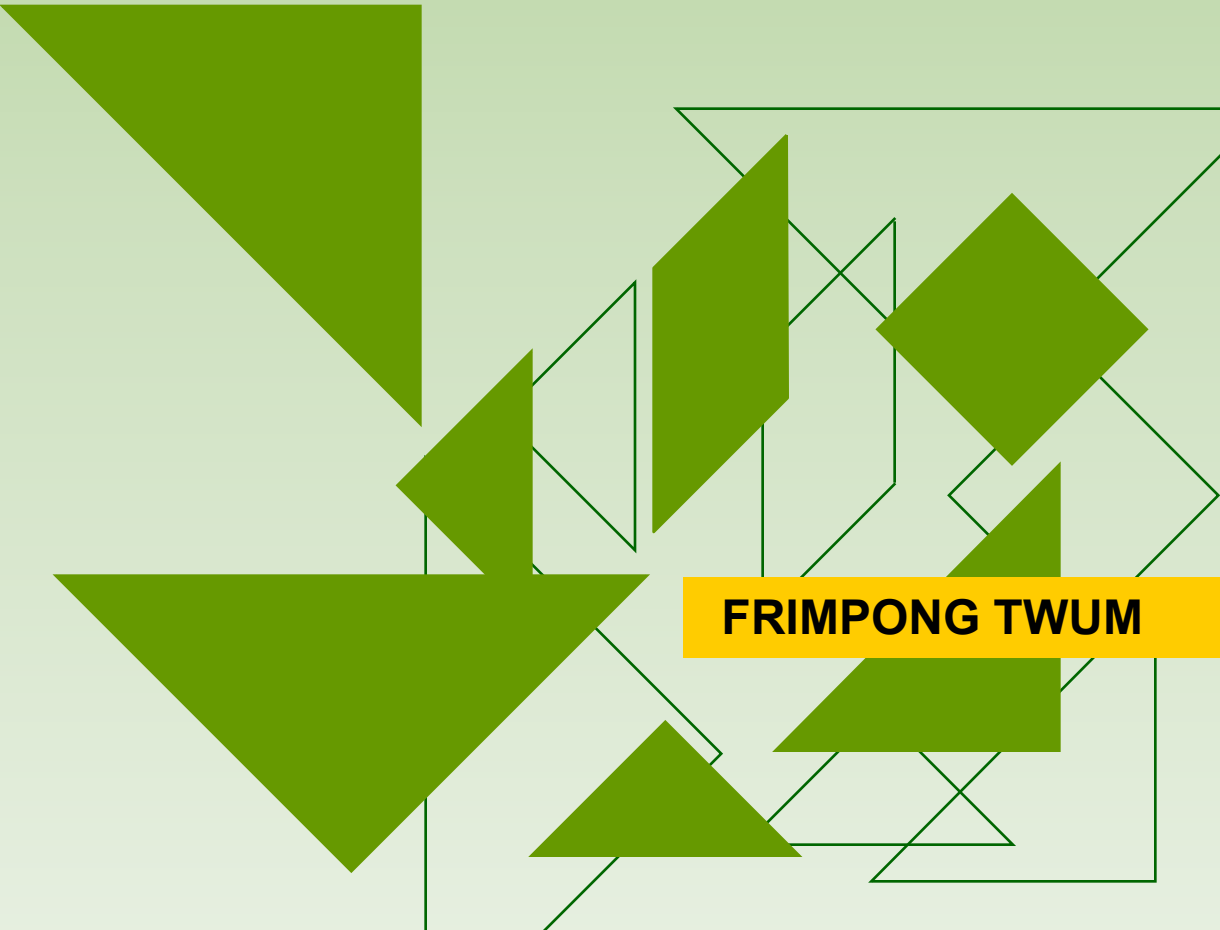


CSM 478 COMPUTER NETWORKS

LECTURE 1: NETWORKING FUNDAMENTALS

An abstract geometric design featuring various green shapes (triangles, diamonds, and polygons) and thin green lines that intersect to form a network-like structure. The design is positioned in the lower right quadrant of the slide.

FRIMPONG TWUM

NETWORK OVERVIEW



What is a Network?

a)

Word-of-Mouth

b)

Telephone



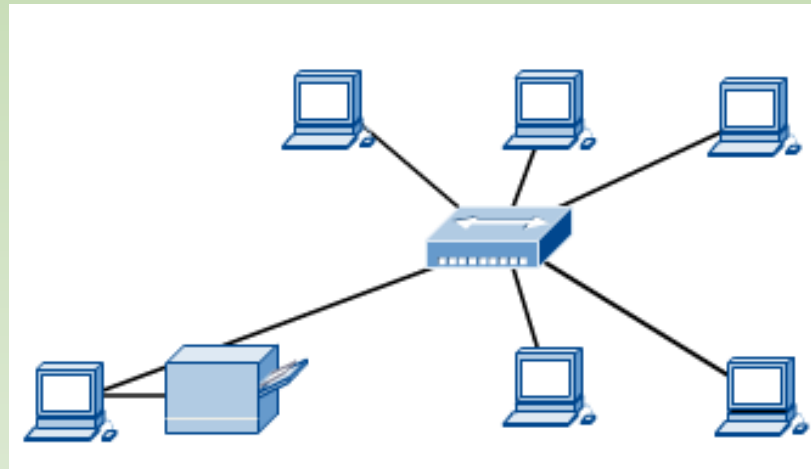
What is Computer Network?

c)

An expansion of (a) and (b) above

What is Computer Network?

- ◆ A computer network is hence, a system of computers, print devices, network devices, and software, linked by communication media.



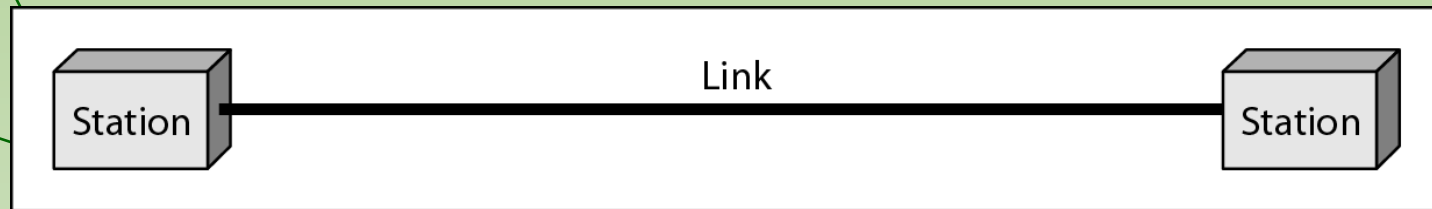
- ◆ A **computer network** is simply two or more computers that are linked together.

What is Computer Network?

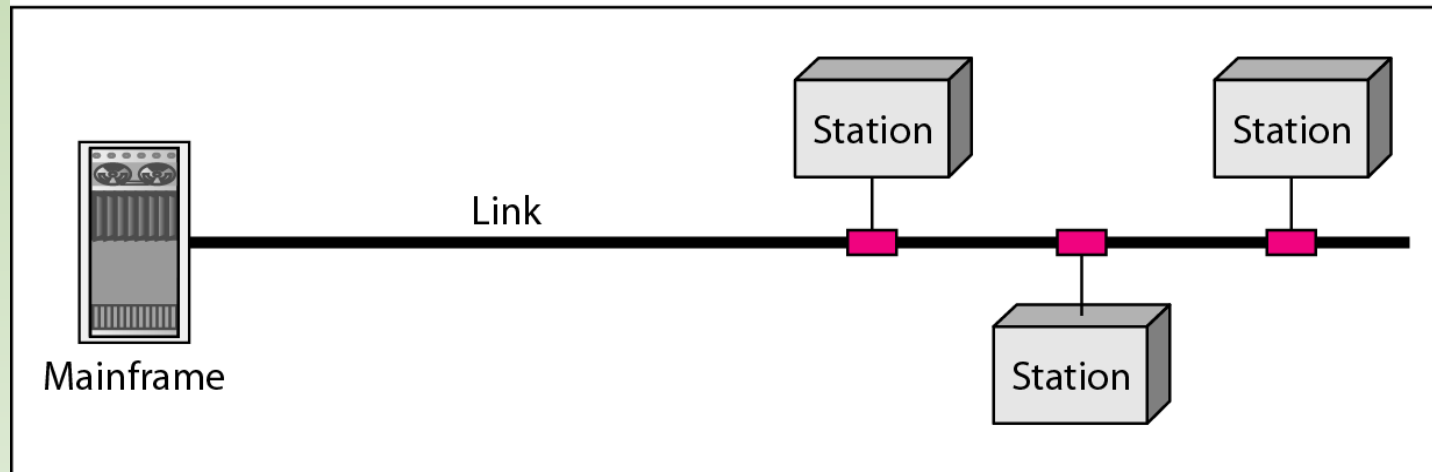
Ctnue...(Some Definitions)

- ◆ A *computer network* is simply two or more computers that are linked together.
- ◆ A *computer network* is defined as having two or more devices (such as workstations, printers, or servers) that are linked together for the purpose of sharing information, resources, or both.
- ◆ A *computer network* is hence a set of devices (often referred to as **nodes**) connected by communication links. A **node** can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Types of connections: point-to-point and multipoint



a. Point-to-point



b. Multipoint

COMPUTER NETWORKS Vrs. DATA COMMUNICATIONS

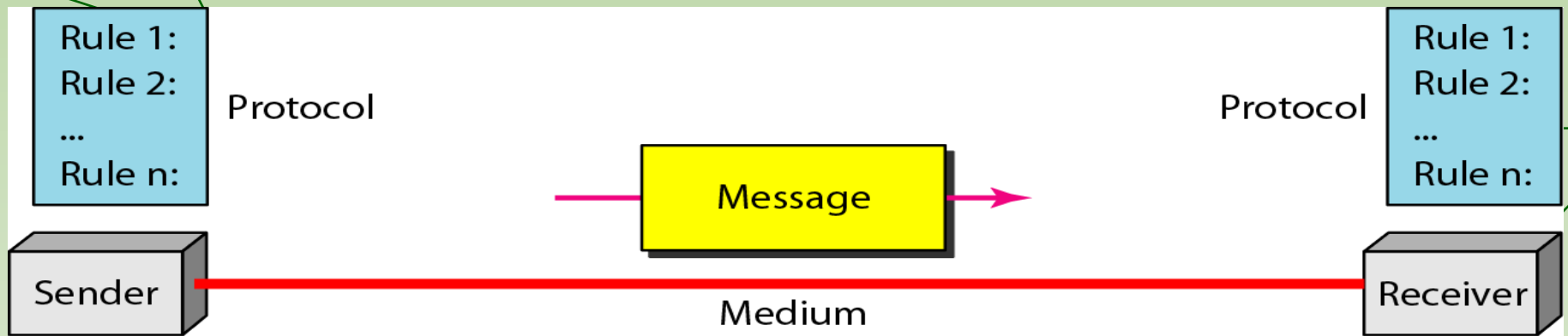
- ◆ *Computer network* exist so that data may be sent from one place to another: (this is the basic concept of ***Data Communications***).

What is Data Communication?

- ◆ The term telecommunication means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- ◆ Computer Networks are the highways on which data travels.

Components of a Data Communications System

- ◆ A Data Communications System has five components as follows:

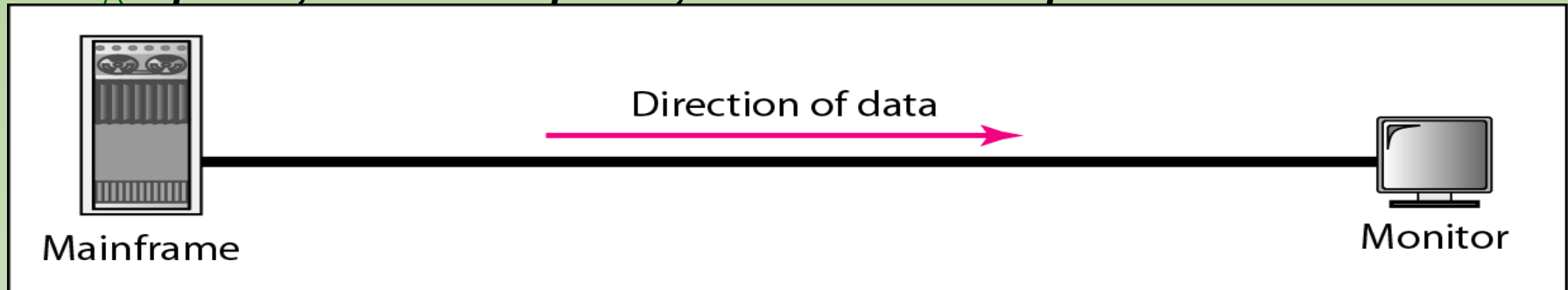


- ◆ The effectiveness of a data communications system depends on 4 fundamental characteristics:

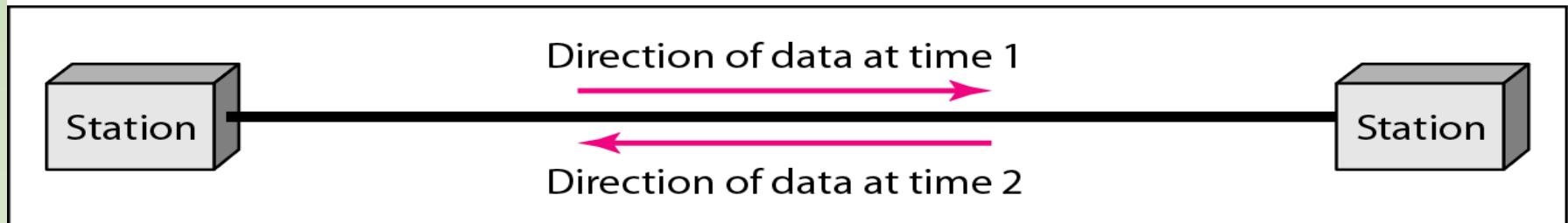
Delivery, Accuracy, Timelines, and Jitter

Data flow (Transmission modes)

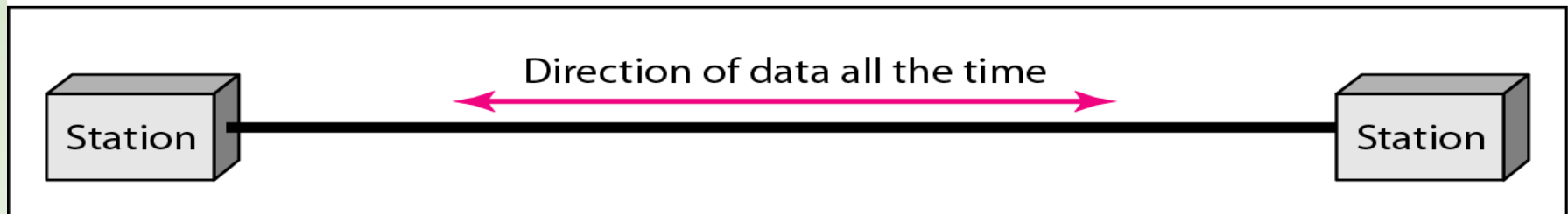
- ◆ Communication between two devices can be *Simplex*, *Half-Duplex*, and *Full-Duplex*



a. Simplex

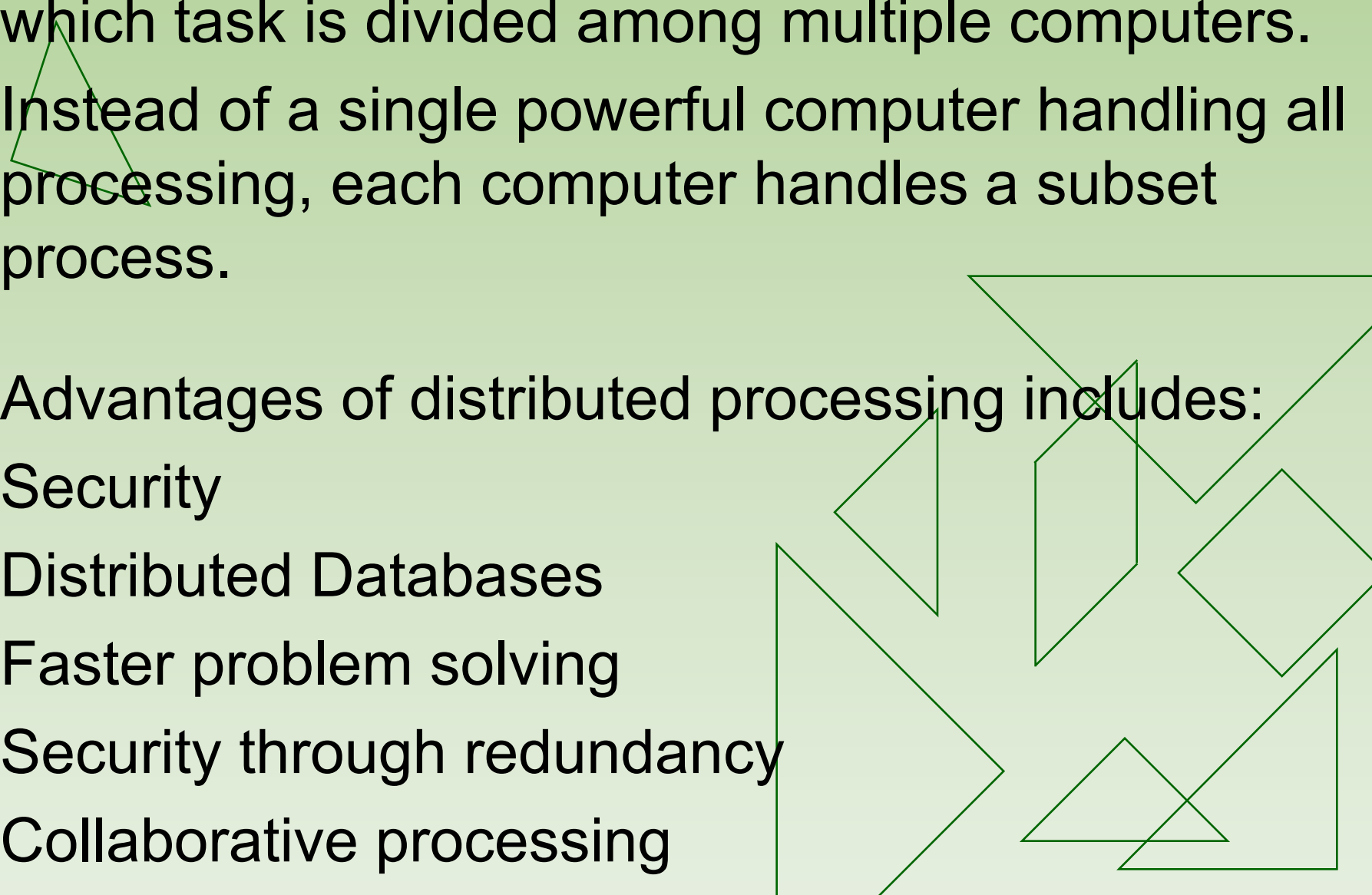


b. Half-duplex



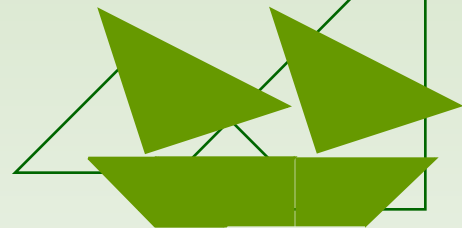
c. Full-duplex

DISTRIBUTED PROCESSING

- ◆ Computer Networks use distributed processing in which task is divided among multiple computers.
 - ◆ Instead of a single powerful computer handling all processing, each computer handles a subset process.
 - ◆ Advantages of distributed processing includes:
 - Security
 - Distributed Databases
 - Faster problem solving
 - Security through redundancy
 - Collaborative processing
- 
- The slide features several decorative green geometric shapes. A small triangle is positioned to the left of the second bullet point. A large, complex shape composed of multiple overlapping triangles and quadrilaterals is located on the right side of the slide, partially overlapping the text of the third bullet point.

BENEFITS OF A NETWORK

- ◆ ***Information sharing:*** Authorized users can use other computers on the network to access and share information and data. This could include special group projects, databases, etc.
- ◆ ***Hardware sharing:*** One device connected to a network, such as a printer or scanner, can be shared by many users.
- ◆ ***Software sharing:*** Instead of purchasing and installing a software program on each computer, it can be installed on the server. All of the users can then access the program from a single location.
- ◆ ***Collaborative environment:*** Users can work together on group projects by combining the power and capabilities of diverse equipment.



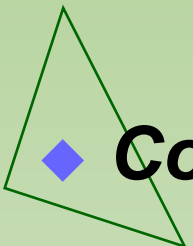
RISKS OF NETWORK COMPUTING

- ◆ The security of a computer network is challenged everyday by:
 - Equipment malfunctions
 - System failures
 - ◆ Note: equipment malfunctions and system failures may be caused by natural disasters such as floods, storms, or fires, and electrical disturbances
 - Computer hackers
 - Virus attacks

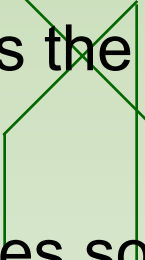
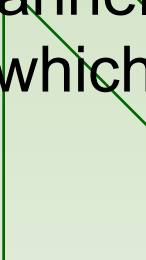
Criteria for assessing a computer network effectiveness and efficiency

- ◆ To be considered effective and efficient, a computer network must meet three most important criteria as follows:
- Performance (measured using Transit Time, Response Time). Depends on Number of users, Type of transmission medium, Hardware, Software
- Reliability (measured by accuracy of delivery, network reliability, frequency of failure, recovery time of network after failure, network readiness in event of catastrophe.
- Security (i.e. Protecting data from unauthorised access and viruses.

COMMUNICATIONS MEDIA

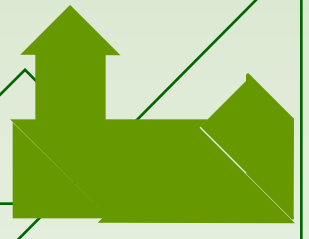


◆ ***Communications Channel***

- To transfer data from one computer to another requires some type of link through which the data can be transmitted. This link is known as the *communications channel*.
 - To send data through the channel requires some type of *transmission media*, which may be either physical or wireless.
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PHYSICAL MEDIA

- ◆ *Twisted-pair cable* – consists of two independently insulated wires twisted around each other (least expensive type of cable—the kind that is used in many telephone systems)
- ◆ *Coaxial cable* – consists of an insulated center wire grounded by a shield of braided wire (the primary type of cabling used by the cable television industry; it is more expensive than twisted pair)
- ◆ *Fiber-optic cable* – contains hundreds of clear fiberglass or plastic fibers (threads) (made from thin, flexible glass tubing; bandwidth is greater, so it can carry more data; it is lighter than metal wires and is less susceptible to interference; it is fragile and expensive)
- ◆ *ISDN line* – a special digital telephone line that transmits and receives information at very high speeds



WIRELESS MEDIA

- ◆ *Microwave system* – transmits data via high-frequency radio signals through the atmosphere
- ◆ *Satellite system* – receive transmitted signals, amplify them, and then transmit the signals to the appropriate locations
- ◆ *Cellular technology* – uses antennae resembling telephone towers to pick up radio signals within a specific area (cell)
- ◆ *Infrared technology* – transmits data as infrared light waves from one device to another, providing wireless links between PCs and peripherals

Note: The type of medium selected is determined by the type of network, the size of the network, and the cost.

TYPES OF NETWORK?

- ◆ Computer Networks are typically classified according to their reach and complexity.
- ◆ The most common types of networks are:
 - ◆ *Local Area Networks (LANs)* and
 - ◆ *Wide Area Networks (WANs)*.
- ◆ The primary difference between the two is that a **LAN** is generally confined to a limited geographical area, whereas a **WAN** covers a large geographical area. Most **WANs** are made up of several connected **LANs**.

TYPES OF NETWORKS

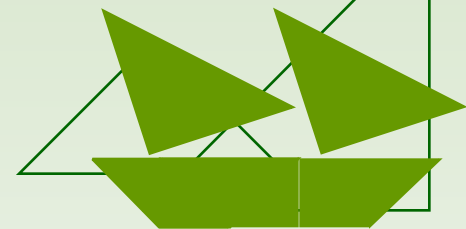
- ◆ **Local Area Network (LAN)** – a network that spans a small area such as a building or an office.

- Software applications and other resources are stored on a **file server**.
- **Print servers enable** multiple users to share the same printer.

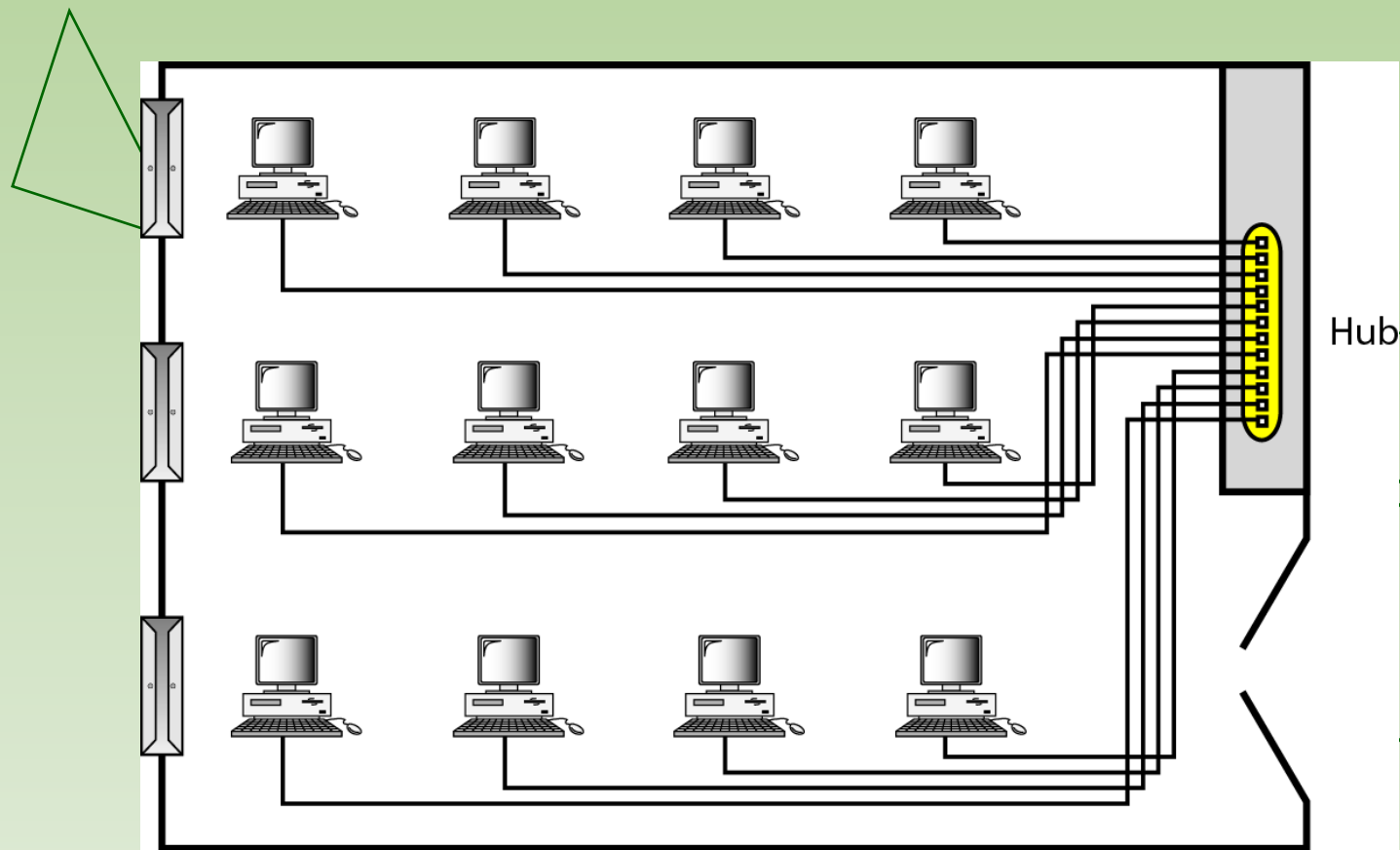
- ◆ **Wide Area Network (WAN)** – a network that spans a wide geographical area; there are several types of WANS

- Metropolitan area network (MAN)
- Public access network (PAN)
- Value added network (VAN)
- Virtual private network (VPN)

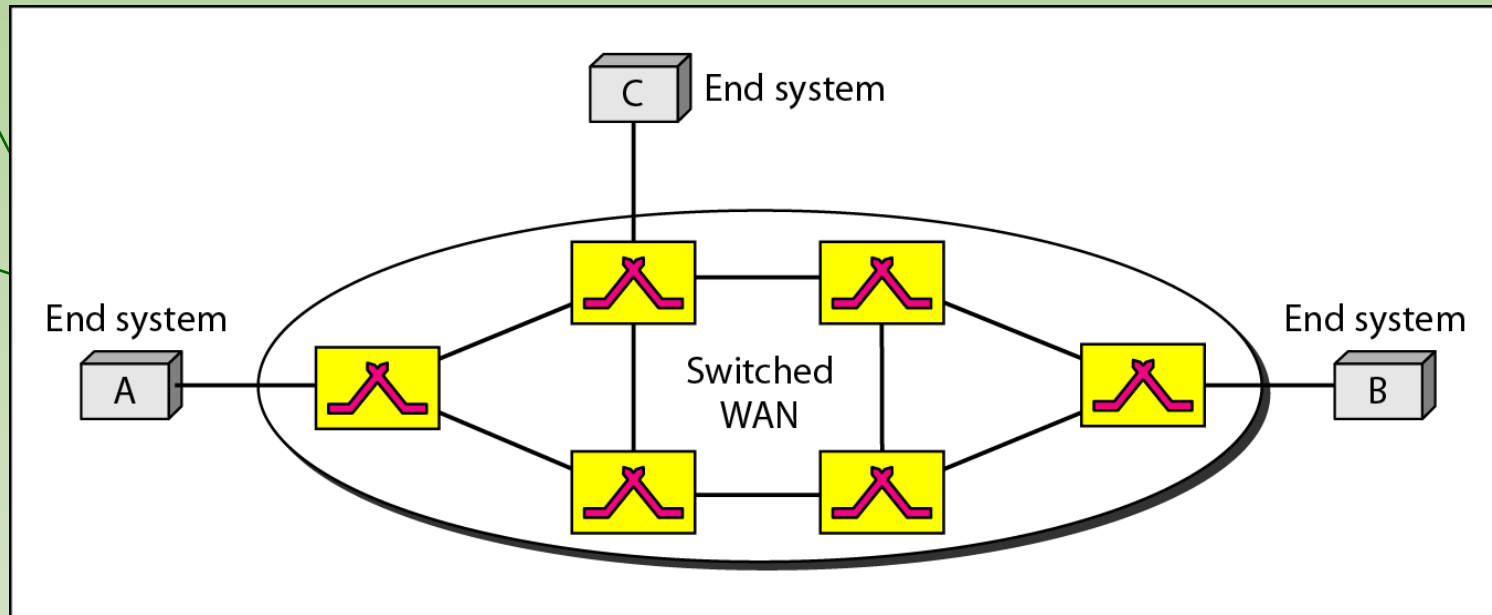
- ◆ **ENTERPRISE NETWORK!!!**



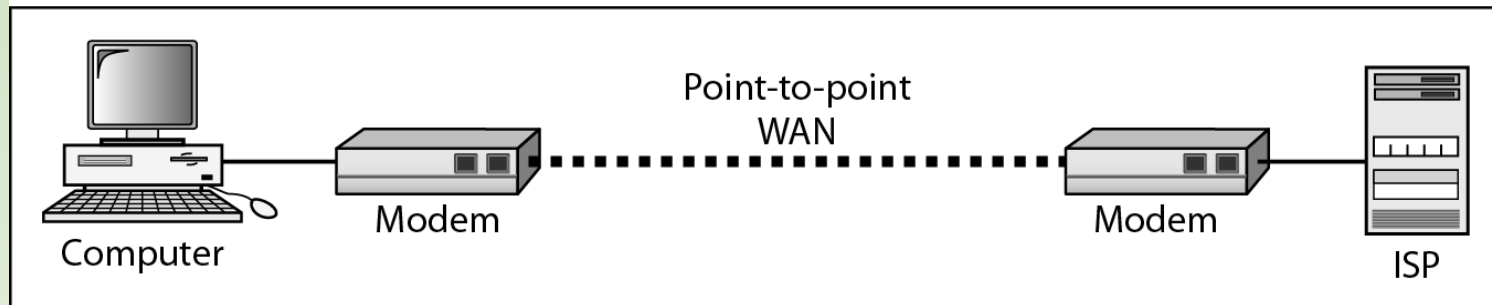
An isolated LAN connecting 12 computers to a hub in a closet



WANs: a switched WAN and a point-to-point WAN

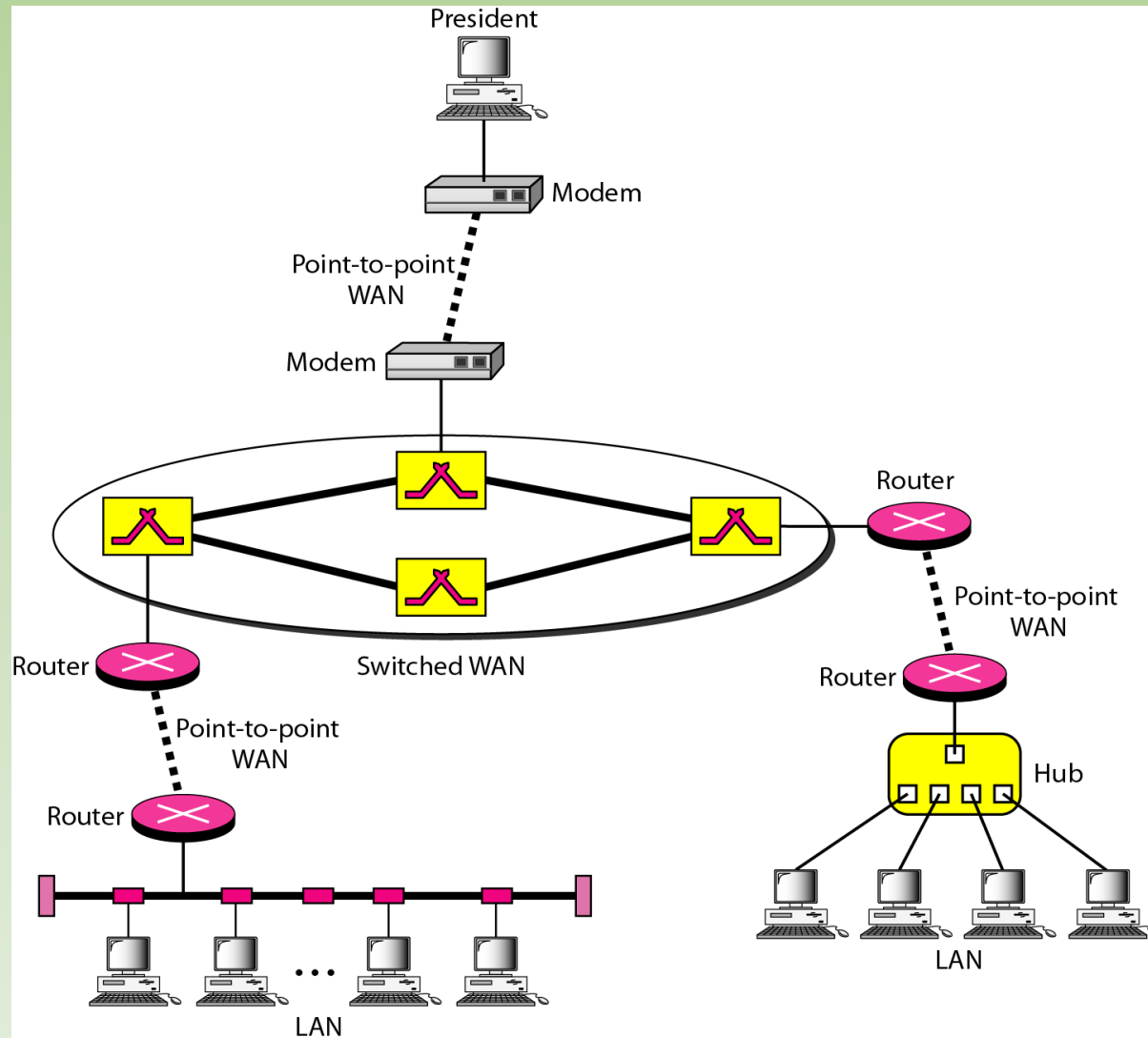


a. Switched WAN



b. Point-to-point WAN

A heterogeneous network made of four WANs and two LANs



TYPES OF NETWORKS

- ◆ **Intranet** – An intranet is a private LAN designed for use by everyone within an organization. An intranet might consist of an internal e-mail system, a message board and one or more Web site portals that contain company news, forms, and personnel information.

Access to an intranet's web site is restricted by a *firewall*.

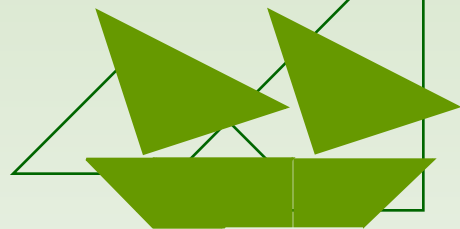
- ◆ **Extranet** – a network that connects people within your company with people who are outside your company--all within a secure, password-protected network that can be accessed from anywhere.

HOW ARE NETWORKS CATEGORIZED?

- ◆ Networks are usually classified using three properties:
 - ◆ *Protocol*
 - ◆ *Architecture*
 - ◆ *Topology*

CATEGORISATION BY NETWORK/COMMUNICATION PROTOCOLS

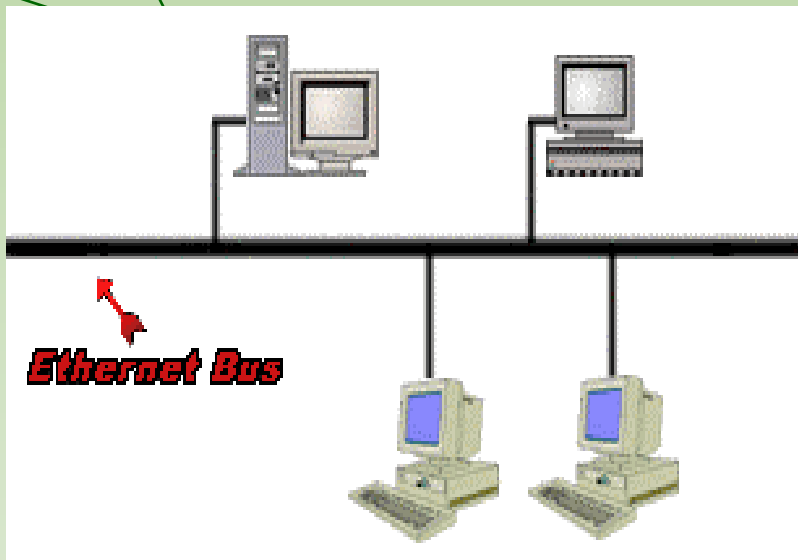
- ◆ ***A protocol*** is simply an agreed-on set of rules and procedures for transmitting data between two or more devices.
- ◆ ***Features determined by the protocol are:***
 - How the sending device indicates it has finished sending the message.
 - How the receiving device indicates it has received the message.
 - The type of error checking to be used.



NETWORK/COMMUNICATIONS PROTOCOLS

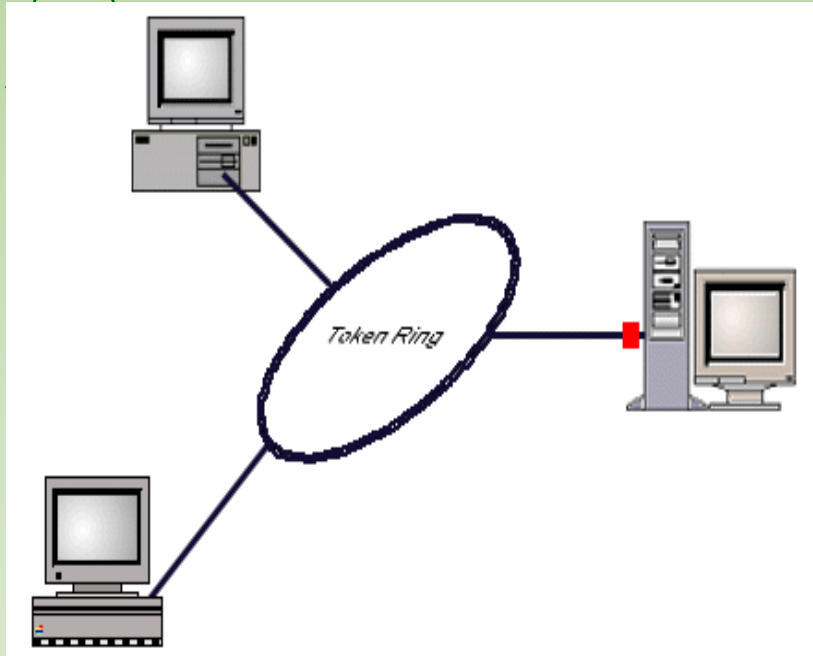
- ◆ NETWORK PROTOCOLS: Most networks use *Ethernet*, but some network may use *IBM's Token Ring* protocol.
- ◆ COMMUNICATIONS PROTOCOLS: On the Internet, the major protocol is ***TCP/IP*** (an acronym for Transmission Control Protocol/Internet Protocol).

EXAMPLE OF AN ETHERNET BUS



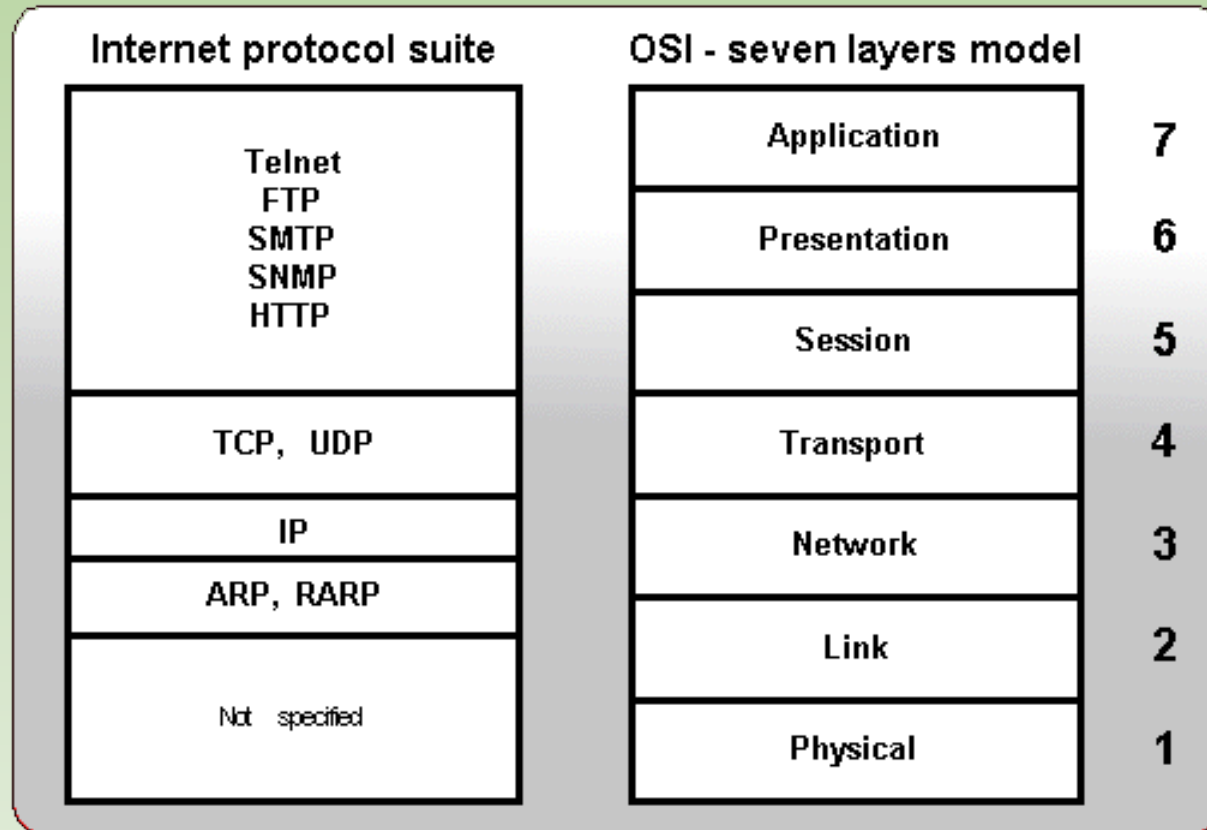
- ◆ Ethernet LANs use a bus topology.
- ◆ All stations are connected to a single long cable.
- ◆ Any station can send a signal along the cable, which all other stations will receive. Unlike ring topologies, the cable doesn't close a loop.

TOKEN RING NETWORK



- ◆ A token passing ring LAN is a group of computers connected in a loop. The group uses a token passing access mechanism.
- ◆ A computer wishing to send data should first receive permission. When it gets control of the network it may transmit a frame. Each frame transmitted on the ring is transmitted from one computer to the next, until it ultimately returns to the initiator of the transmission.

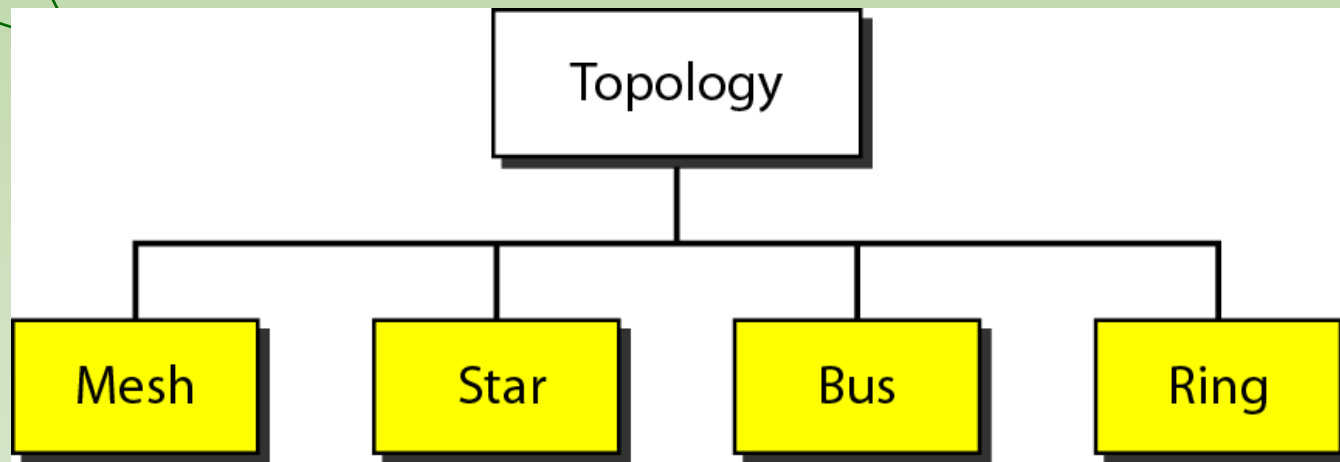
INTERNET PROTOCOLS



CATEGORISATION BY NETWORK ARCHITECTURE

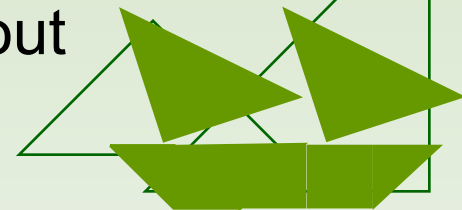
- ◆ *Network architecture* – refers to the way a network is designed and built. The two major types are:
 - *Peer-to-peer architecture* – Computers connect with each other in a workgroup to share files, printers, and Internet access. This is used to connect a small number of computers.
 - *Client/server architecture* – sends information from a client computer to a server, which then relays the information back to the client computer, or to other computers on the network

CATEGORISATION BY NETWORK TOPOLOGY

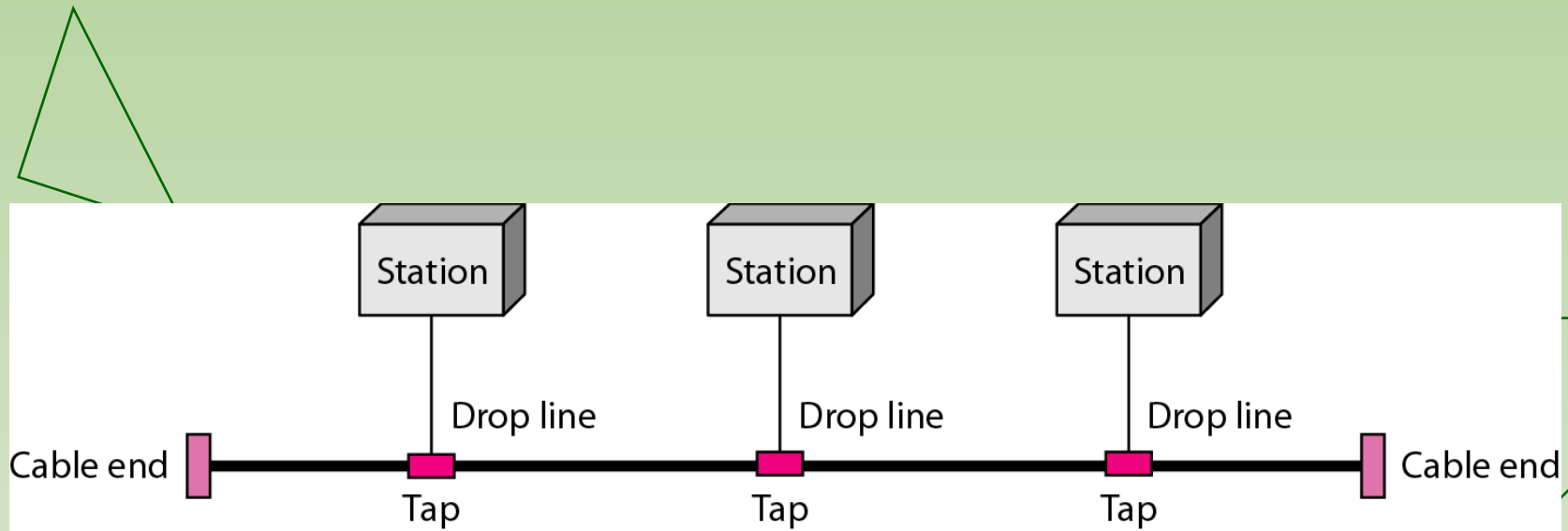


NETWORK TOPOLOGIES

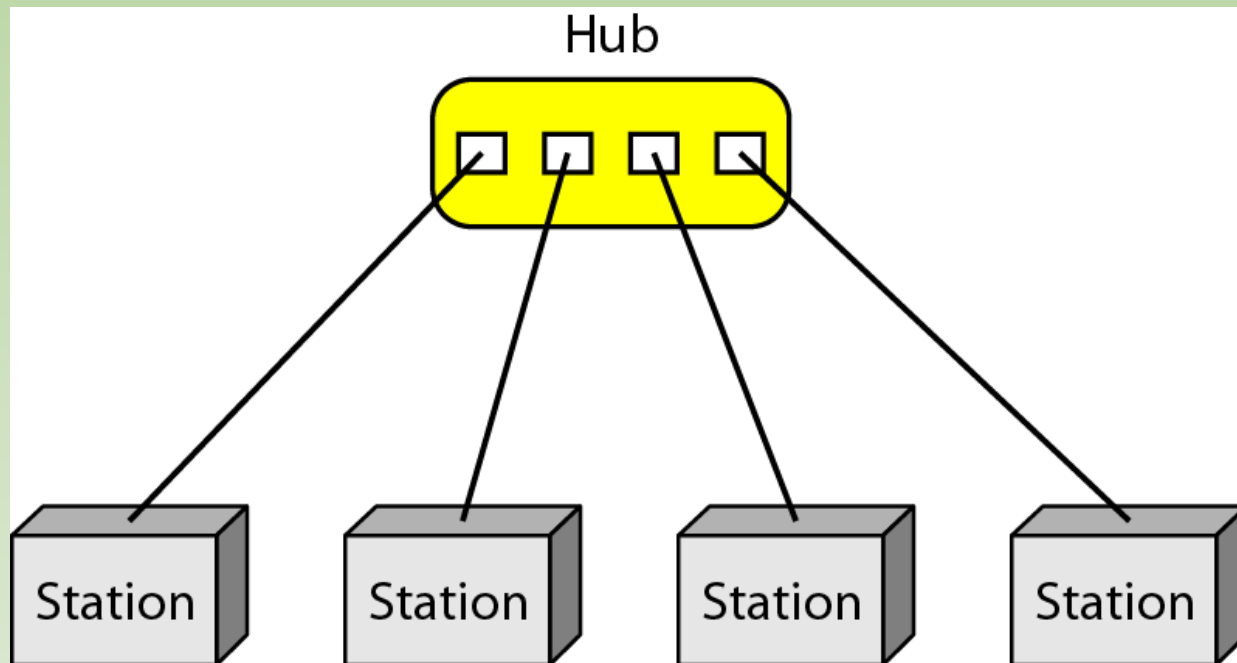
- ◆ *Network Topology (or layout)* – refers to the way computers and peripherals are configured to form networks
 - *Bus topology* – all computers are linked by a single line of cable
 - *Star topology* – multiple computers and peripheral devices are linked to a central computer, called a *host*
 - *Ring topology* – each computer or workstation is connected to two other computers, with the entire network forming a circle
 - *Hybrid topology* – combine network layout types to meet their needs



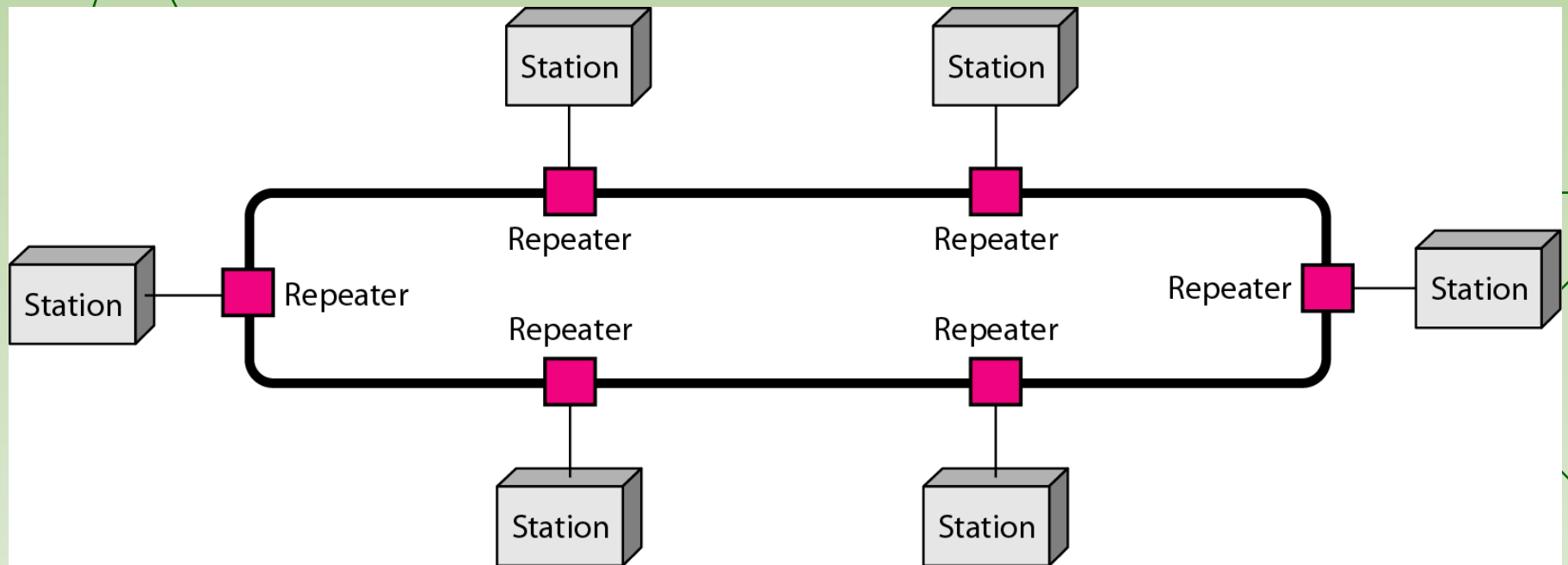
A bus topology connecting three stations



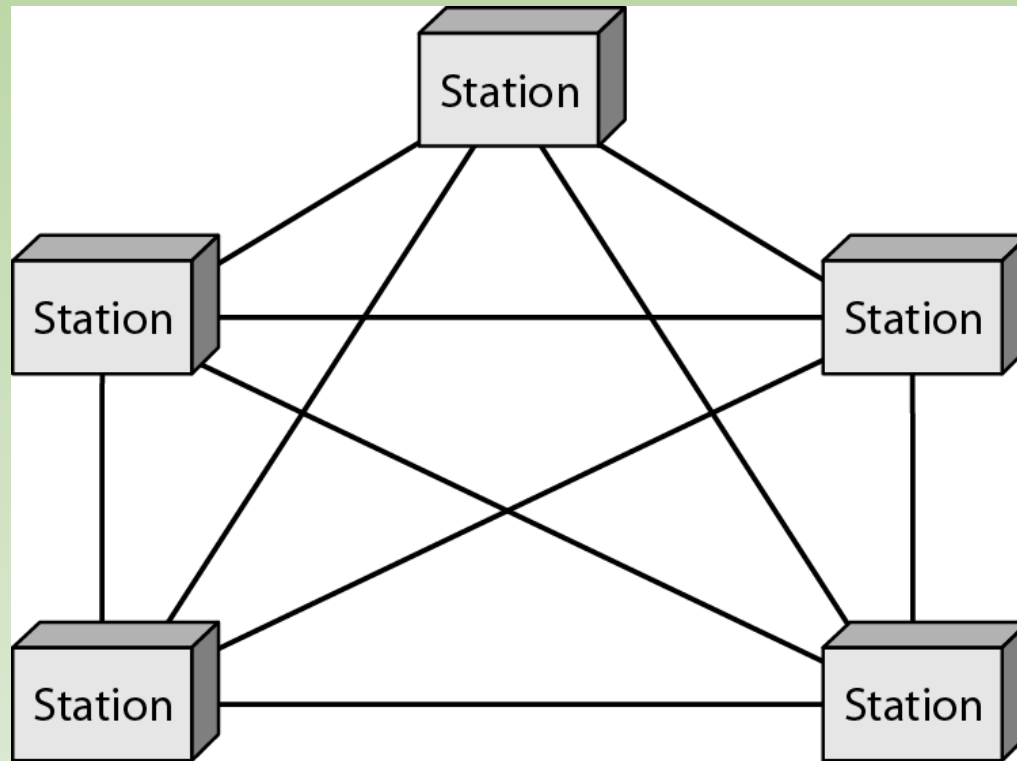
A star topology connecting four stations



A ring topology connecting six stations



A fully connected mesh topology (five devices)



A hybrid topology: a star backbone with three bus networks

